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System Analysis and Design Report : Adding an Enhanced Feature to the NewPoint Accounting System for Al-Nahdah Auto Parts Company.

Dr.Arwa Mohammed Asiri

1ST SEMESTER 2026

New Point

Prepared by:

[redacted]

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Executive Summary

This report provides a comprehensive analysis and redesign proposal for the NewPoint accounting system, aiming to enhance operational efficiency, improve system performance, and strengthen supplier communication. Through a data collection process—including employee interviews, questionnaires, and document analysis—key issues impacting the current system were identified. Key challenges include manual inventory tracking, inefficient supplier communication, and purchase order delays.

Customers and employees reported issues such as stock shortages, slow processing times, and a lack of automation in the interaction with suppliers. These problems contribute to delays in restocking inventory and overall inefficient workflows. Based on these findings, a comprehensive set of functional and non-functional requirements was developed.

The system will be addressed in ways that include automating inventory tracking and supplier notifications to ensure timely restocking; improving system speed to handle purchase orders more efficiently; and implementing a unified platform for comparing supplier quotes to streamline decision-making.

The analysis requires the creation of system models, such as context diagrams, DFDs, and child diagrams, for both the current and proposed systems. MoSCoW technology was used to prioritize requirements, ensuring the delivery of key features in Sprint 1. These improvements will streamline processes, enhance the user experience, and increase customer confidence.

The successful implementation of these system enhancements will not only improve daily operations but also help Al Nahda Auto Parts regain supplier trust, strengthen its competitive advantage, and ensure sustainable long-term business growth.

4.1 Introduction

This chapter describes several methods used to gather essential data on improvements and suggestions for the New Point accounting system used by Al Nahda Auto Parts Company. Various data collection techniques were employed to understand the company's challenges and system limitations.

Several interviews were conducted with users, including inventory managers, accountants, and the system developer, to identify weaknesses and problems such as manual communication with suppliers, delayed purchasing cycles, and inventory tracking issues. Each interview question was linked to project objectives to ensure accurate results.

Furthermore, the chapter includes a detailed analysis of data and documentation, such as purchase records and reports, to validate workflows. This process identified system requirements to ensure the solution aligns with needs and achieves objectives.

Chapter 4: Requirement Elicitation

4.2 Interview

4.2.1 Interview Preparation

To ensure the effectiveness and accuracy of the interview process, thorough preparation was conducted before implementation.
The primary objectives :
1-To understand employee satisfaction with the current purchasing and supplier-interaction process
2-To evaluate the efficiency and speed of the current purchasing and inventory-tracking process
3-To develop the system and assess employees' understanding and usage of current NewPoint features

The first step in the preparation process involved selecting a representative **sample of participants** from various roles within the company. Five employees were chosen, including an inventory manager, an accountant, a system operator, and two sales staff members. This diversity ensured that the feedback reflected the actual operational challenges across different departments.

The interview questions were designed in a **funnel-like fashion**, starting with open-ended questions to encourage detailed responses and gradually progressing to closed-ended questions to elicit more specific input. This structure helped maintain the interview's focus, organization, and effectiveness in gathering relevant information. All questions were carefully aligned with the project objectives, and provides an easy, nonthreatening way to begin an interview.

The interviews were scheduled over **three days**, **from Sunday, March 24, 2025, to Tuesday**, **March 26, 2025**, and were conducted in the **evenings between 8:00 and 9:00 PM via Zoom** to accommodate participants' work schedules. Each participant was informed in advance of the purpose of the interview, its expected duration, and the guarantee of confidentiality. Prior to the interviews, a consent form and an information sheet were prepared and distributed electronically to ensure all participants fully understood their rights and the scope of the study.

4.2.2 Interview questions objectives mapping

Objectives	Questions
To understand employee satisfaction with the current purchasing and supplier-communication process	1-Can you describe your experience with the current purchasing workflow in the NewPoint system?
	2-Have you ever faced issues when contacting suppliers manually? If yes, how did these issues affect your work?
	3-What challenges or frustrations do you experience when tracking inventory shortages or communicating with suppliers?
To evaluate the efficiency and speed of the current purchasing and inventory-tracking process	1-Have you ever experienced delays when discovering stock shortages? If yes, how were you notified?
	2-How would you describe the speed and efficiency of comparing supplier quotations manually?
	3-Have manual communication or late responses from suppliers affected purchase decisions or created pressure in your workflow?
To develop the system and assess employees' understanding and usage of current NewPoint features	1-What features in the NewPoint system do you use most when creating purchase orders, and why are they useful?
	2-Are there any parts of the system that you find unclear or require extra manual effort?
	3- What new features do you wish the NewPoint system had to make your job easier?

Table 4.1: Interview Questions and Objectives Mapping

4.2.3 Information sheet and consent form

Interview Information Sheet

a copy for the participant

Study Purpose:
The objective is to gather information to improve Al Nahda Auto Parts' New Point accounting system, particularly in the areas of inventory management, supplier communication, and purchase order processing.

Interview Purpose:
As a team of systems analysts working to improve Al Nahda Auto Parts' New Point system, we are conducting interviews with employees to understand their daily challenges, needs, and expectations. Your feedback will help identify system limitations and guide the development of new features that improve automation, efficiency, and overall system usability.

What will you be asked to do?
You will be asked several questions about your experience with the New Point system, including tasks related to inventory, supplier communication, and purchase orders. The interview will last approximately 15-20 minutes and will be recorded for internal analysis purposes only.

Confidentiality:
Your answers will remain confidential. No personal information, such as your name, phone number, or email address, will be recorded or shared. All information collected will be used exclusively for system improvement purposes.

Data Usage:
Interviewers will use the information and statements provided by participants exclusively to analyze system performance and develop proposed improvements for Al Nahda Automotive Spare Parts.

Data Access:
Access to collected data will be available only to interviewees. No information will be shared with third parties.

Data Storage:
All data will be stored securely in accordance with the organization's privacy policies and data protection laws.

Interviewer Contact Information:
Reema Jabakji - 0568109014
Sumiah Bawazir - 0590555497
Maha Saleh - 0532290985

Figure 4.2: Participant Information Sheet

Conset Form

1- Type of the interview:

Individual

2- Target participants:

Employees of Al-Nahdah Auto Parts who use the NewPoint Accounting System

3- Context:

March 24, 2025 – Monday
At 8:00 PM
Online via Zoom

4- Participants' rights:

- You have the right to choose whether or not to participate. Participation is entirely voluntary.
- You can stop or withdraw from the interview at any time without any consequences.
- You can skip any question you do not wish to answer.
- Your answers will remain confidential, and your identity will not be revealed or published.

5- Agreement:
This confirms your consent to participate in this study. I have read the information and consent form provided, and I understand its contents.

6- Confirmation:
I confirm that I am over 18 years of age and that I have read and understood the information in this form. I voluntarily consent to participate in this interview and agree that my responses may be used for system analysis and improvement purposes.

Signature: _____
Name: _____
Date: _____
Email: _____

Figure 4.3: Consent Form

4.2.4 Interview Analysis

After conducting employee interviews and analyzing their feedback, several recurring issues affecting the system and its efficiency were identified. The most prominent issues include:

1. Manual Inventory Management and Supplier Communication: Employees reported that this process is time-consuming and prone to errors, causing delays in order fulfillment.

2. Lack of Automation: The absence of a system for automatically notifying employees of stock shortages or generating orders leads to delays in restocking processes.

3. Supplier Communication: Communicating with suppliers through multiple channels (such as email or phone calls) creates confusion and makes it difficult to compare offers.

4. User Interface: Some employees noted that the current system is difficult to navigate, slowing down workflow.

When asked about the impact of these issues, employees expressed frustration and work overload, which negatively affects their efficiency and the quality of service provided to customers.

To address these problems, the following solutions are recommended:

1. Automate inventory management with automatic supplier notifications.

2. Implement a unified platform for suppliers to submit offers and compare prices.

3. Improve system speed for increased efficiency.

4. Update the system interface for a simpler and more user-friendly experience.

5. Add live chat for easier and more immediate communication.

Implementing these modifications will improve efficiency, reduce errors, boost productivity, and increase customer satisfaction.

4.3 Questionnaire

4.3.1 Questionnaire Preparation

The second data collection method used in this study was a questionnaire. This questionnaire targeted employees of Al Nahda Auto Parts Company who use the NewPoint accounting system.

The **objectives**:

Objective 1: To measure employees' satisfaction with the current inventory and supplier-interaction processes.

Objective 2: To evaluate employees' readiness and openness toward adopting automated features within the system.

objective 3: To identify which system enhancements (e.g., automated notifications, price-quote comparison, unified communication) are most valued by users.

Objective 4: To gather quantitative interval data on usability issues related to system speed

The first step was to define the main objectives of the questionnaire, such as assessing the system's ease of use, its effectiveness in inventory management, the speed of purchase order processing, and overall employee satisfaction. The questionnaire was carefully designed, starting with general demographic questions, followed by closed-ended questions to elicit clear opinions about the NewPoint system. Finally, we added open-ended questions to allow employees to share their additional comments or suggestions.

The questionnaire was distributed via internal WhatsApp groups and email to ensure the participation of all targeted employees. This tool helped us gather structured opinions and feedback directly from system users, enabling us to identify areas for improvement and enhance the overall efficiency of the NewPoint system.

4.3.2 Questionnaire Questions objectives mapping

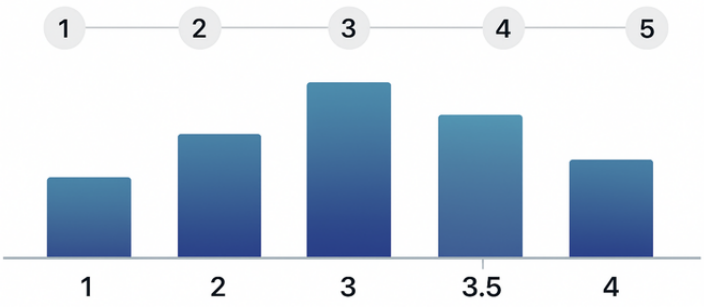
Objectives	Questions
Objective 1: To measure employees' satisfaction with the current inventory and supplier-interaction processes.	On a scale from 1 to 5, how satisfied are you with the current inventory process in the NewPoint system? (1 = Strongly dissatisfied, 5 = Strongly satisfied)
	Which part of the supplier-interaction process do you find most challenging? (Receiving quotes / Updating supplier info / Communication / None)
Objective 2: To evaluate employees' readiness and openness toward adopting automated features within the system.	to what extent do you agree with adopting automated features when interacting with suppliers? (agree, disagree, neutral)
	Are you open to using automated alerts and notifications within the system? (Yes / No)
To identify which system enhancements (e.g., automated notifications, price-quote comparison, unified communication) are most valued by users.	Which supplier-interaction features would be most valuable for you in the system? A. Automated notifications and reminders to suppliers when shortage happens B. A comparison view for suppliers' price quotes C. In-system messaging or communication logs D. All of the above
Objective 4: To gather quantitative interval data on usability issues related to system speed	How satisfied are you with the purchase order processing speed? Neutral / dissatisfied/ very dissatisfied/ satisfied / Very satisfied

Table 4.4: Questionnaire Questions and Objectives Mapping

4.3.3 Questionnaire Analysis

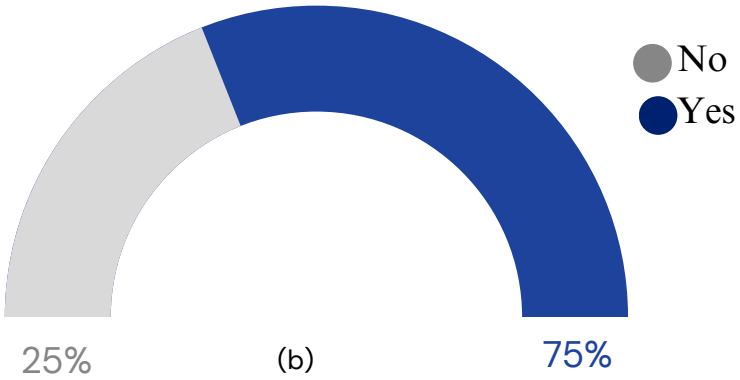
Sum Of Respondents **75**

Employees' satisfaction with the current inventory and supplier-interaction processes.from 1 -5



(a)

openness toward adopting automated features within the system.



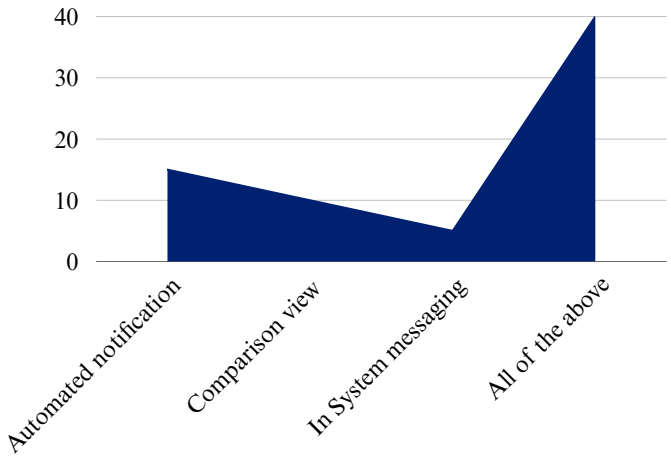
(b)

adopting automated features when interacting with suppliers

Agree	51
Disagree	7
Neutral	17

(C)

The most valuable system enhancement



(d)

Satisfaction with the purchase order processing speed

Satisfaction	1,000 and under	2,000 - 5,000	6,000 - 9,000	10,000 and above	Total
Neutral	6	9	8	7	30
Dissatisfied	3	4	3	2	12
Very dissatisfied	2	3	2	1	8
Satisfied	3	7	4	3	17
Very satisfied	2	—	3	3	8

(e)

In figure (A), it shows the distribution of employee satisfaction levels regarding the current inventory and supplier-interaction processes in the NewPoint system measured on a 1-5 scale. The major part of the responses is located between 3 and 3.5, which means that most of the employees feel moderately satisfied. A handful of people went for the lower scores like 2.5, while only a small number of users marked 4 and 5 at the higher end of the scale for their satisfaction. On the whole, the conclusions indicate that although satisfaction is normally at an average level, there is still a noticeable need for improvement with regard to the efficiency and the user-friendliness of the system.

In Figure (B), a gauge chart illustrates the level of acceptance of the automated features by the employees. According to the findings, 75% of the participants were in favor of the system saying a clear "Yes," meaning they were very ready for process automation to take place. On the other hand, 25% did not think so and said "No," which means they were somewhat reluctant to have automation in the system. The general tendency of the employees has been pointed out as a positive one concerning the switch to more automated workflows.

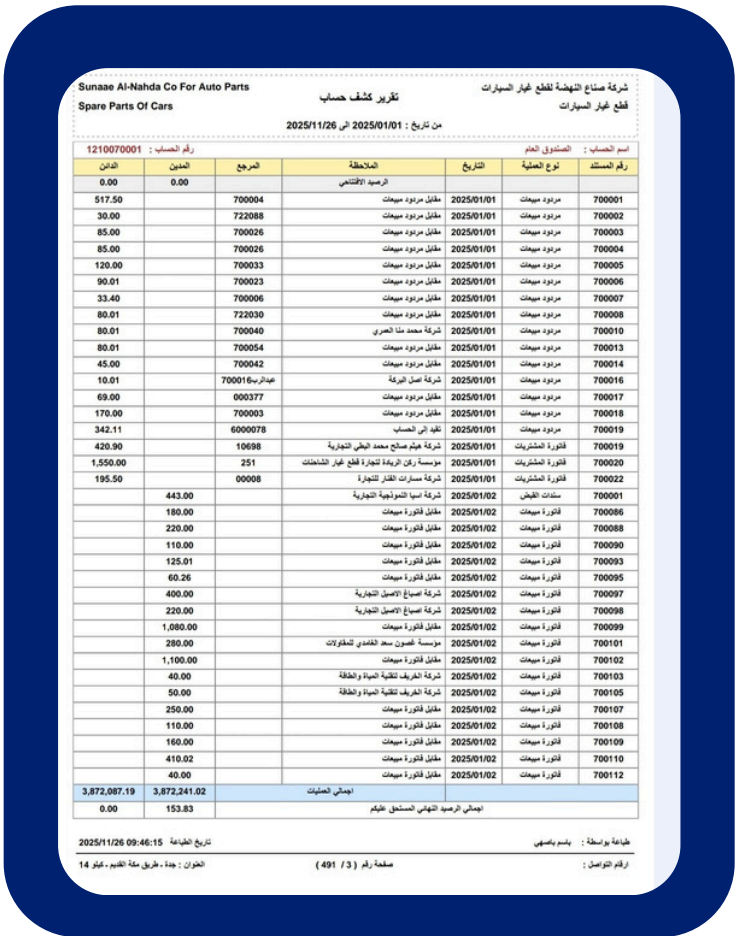
Figure (C), gives an overview of the positions of the employees concerning the automated features in supplier-interaction activities. 51 employees out of 75 respondents were in favor, 17 were neutral, and just 7 were against. This distribution makes it very clear that the majority of employees support automating processes especially such as getting quotes, sending updates, and making communication easier.

Figure (D) shows employees' preferences regarding which system enhancements they find most valuable. The results indicate that the option "All of the above" received the highest number of selections, highlighting a strong desire for a combination of improvements. These include automated notifications, a comparison view for supplier quotes, and in-system messaging. The individual features received lower but varying levels of interest, confirming that employees prefer an integrated enhancement rather than a single isolated feature.

Finally, in Figure (E), we observed the distribution of satisfaction levels with purchase order processing speed across different income brackets. A significant proportion of users, particularly those with middle incomes, reported neutral or moderate dissatisfaction. This suggests that while the speed is acceptable to some employees, others feel the process takes longer than expected. The system may require minor backend improvements rather than a complete redesign.

In addition to the graphs, employees were also asked about improvements they would like to see in NewPoint. The most common responses included more user-friendly interface elements, simplified navigation, and faster load times, especially during purchase order creation. These improvements would enhance daily operations and increase overall satisfaction with the NewPoint system.

4.4 Documents Analysis



شركة صناع النهضة لقطع غيار السيارات
قطع غيار السيارات

Sunaae Al-Nahda Co For Auto Parts
Spare Parts Of Cars

تقرير كشف حساب

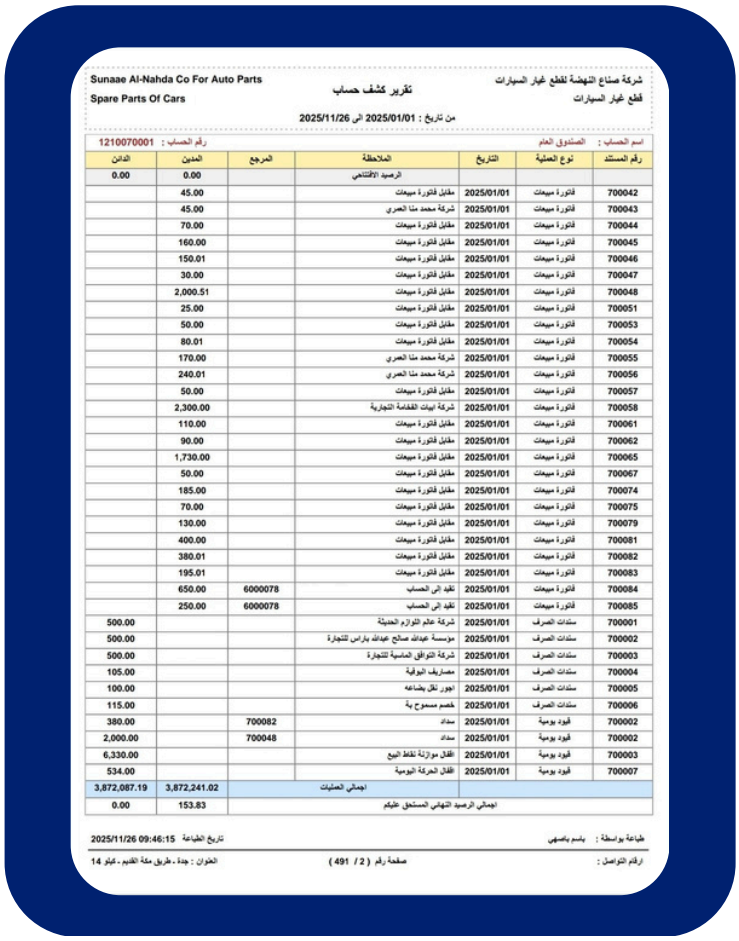
من تاريخ : 2025/01/01 الى 2025/11/26

اسم الحساب : الصندوق العام — رقم الحساب : 1210070001

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طباعة بواسطة : باسم باصهبي — تاريخ الطباعة : 2025/11/26 09:46:15

ارقام التواصل : — صفحة رقم (3 / 491) — العنوان : جدة - طريق مكة القديم - كيلو 14



شركة صناع النهضة لقطع غيار السيارات
قطع غيار السيارات

Sunaae Al-Nahda Co For Auto Parts
Spare Parts Of Cars

تقرير كشف حساب

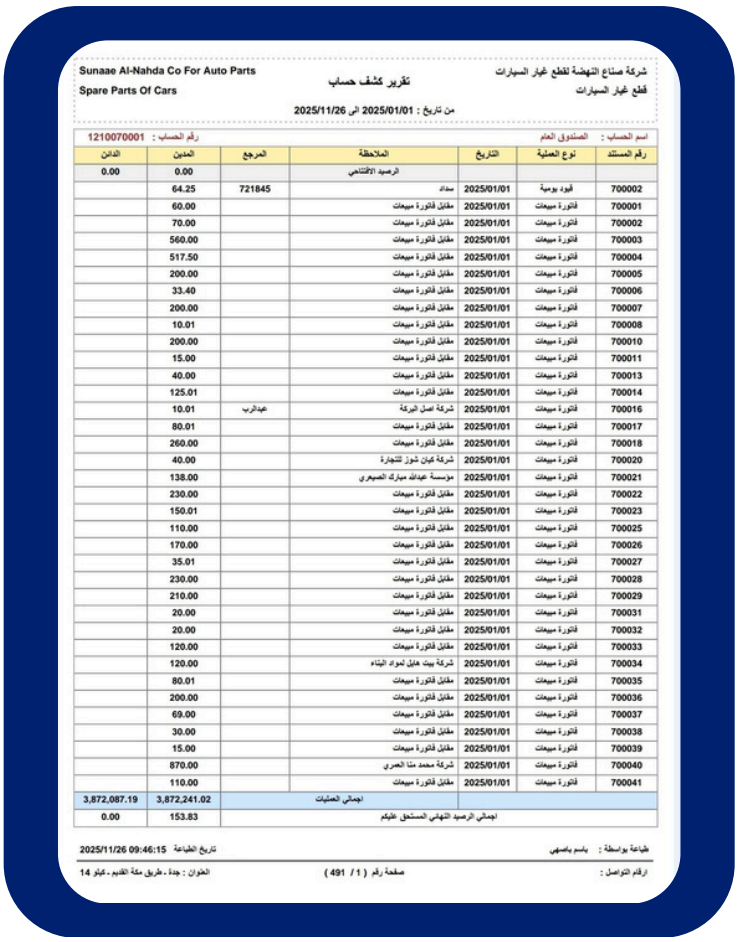
من تاريخ : 2025/01/01 الى 2025/11/26

اسم الحساب : الصندوق العام — رقم الحساب : 1210070001

رقم المستند	نوع العملية	التاريخ	الملاحظة	المرجع	المدين	الدائن
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700084	فاتورة مبيعات	2025/01/01	نقيد الى الحساب	6000078	650.00	
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700001	سندات الصرف	2025/01/01	شركة عالم التوازم الحديثة			500.00
700002	سندات الصرف	2025/01/01	مؤسسة عبدالله صالح عبدالله باراس للتجارة			500.00
700003	سندات الصرف	2025/01/01	شركة التوافق العالمية للتجارة			500.00
700004	سندات الصرف	2025/01/01	مصاريف البوفية			105.00
700005	سندات الصرف	2025/01/01	اجور نقل بضاعه			100.00
700006	سندات الصرف	2025/01/01	خصم مسموح به			115.00
700002	قيود يومية	2025/01/01	سداد	700082		380.00
700002	قيود يومية	2025/01/01	سداد	700048		2,000.00
700003	قيود يومية	2025/01/01	اقفال موازنة نقط البيع			6,330.00
700007	قيود يومية	2025/01/01	اقفال الحركة اليومية			534.00
			اجمالي العمليات		3,872,241.02	3,872,087.19
			اجمالي الرصيد النهائي المستحق عليكم		153.83	0.00

طباعة بواسطة : باسم باصهبي — تاريخ الطباعة : 2025/11/26 09:46:15

ارقام التواصل : — صفحة رقم (2 / 491) — العنوان : جدة - طريق مكة القديم - كيلو 14



شركة صناع النهضة لقطع غيار السيارات
قطع غيار السيارات

Sunaae Al-Nahda Co For Auto Parts
Spare Parts Of Cars

تقرير كشف حساب

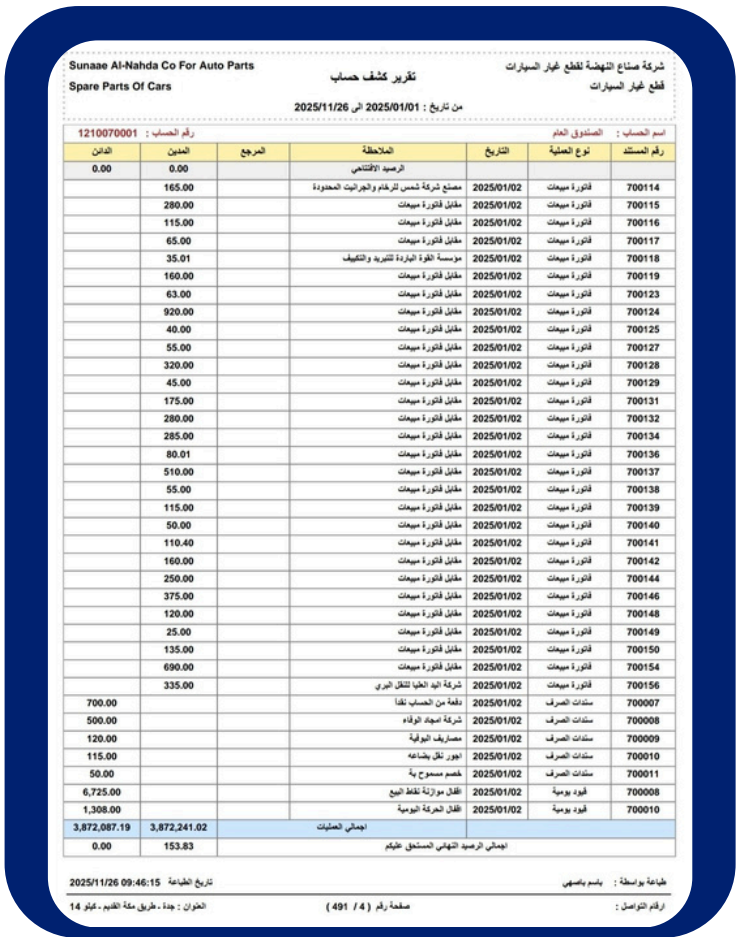
من تاريخ : 2025/01/01 الى 2025/11/26

اسم الحساب : الصندوق العام — رقم الحساب : 1210070001

رقم المستند	نوع العملية	التاريخ	الملاحظة	المرجع	المدين	الدائن
			الرصيد الافتتاحي		0.00	0.00
700002	قيود يومية	2025/01/01	سداد	721845	64.25	
700001	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		60.00	
700002	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		70.00	
700003	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		560.00	
700004	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		517.50	
700005	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		200.00	
700006	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		33.40	
700007	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		200.00	
700008	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		10.01	
700010	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		200.00	
700011	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		15.00	
700013	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		40.00	
700014	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		125.01	
700016	فاتورة مبيعات	2025/01/01	شركة اصل البركة	عبدالرب	10.01	
700017	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		80.01	
700018	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		260.00	
700020	فاتورة مبيعات	2025/01/01	شركة كيان شوز للتجارة		40.00	
700021	فاتورة مبيعات	2025/01/01	مؤسسة عبدالله مبارك الضيعري		138.00	
700022	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		230.00	
700023	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		150.01	
700025	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		110.00	
700026	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		170.00	
700027	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		35.01	
700028	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		230.00	
700029	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		210.00	
700031	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		20.00	
700032	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		20.00	
700033	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		120.00	
700034	فاتورة مبيعات	2025/01/01	شركة بيت هايل لمواد البناء		120.00	
700035	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		80.01	
700036	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		200.00	
700037	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		69.00	
700038	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		30.00	
700039	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		15.00	
700040	فاتورة مبيعات	2025/01/01	شركة محمد منا الحربي		870.00	
700041	فاتورة مبيعات	2025/01/01	مقابل فاتورة مبيعات		110.00	
			اجمالي العمليات		3,872,241.02	3,872,087.19
			اجمالي الرصيد النهائي المستحق عليكم		153.83	0.00

طباعة بواسطة : باسم باصهبي — تاريخ الطباعة : 2025/11/26 09:46:15

ارقام التواصل : — صفحة رقم (1 / 491) — العنوان : جدة - طريق مكة القديم - كيلو 14



شركة صناع النهضة لقطع غيار السيارات
قطع غيار السيارات

Sunaae Al-Nahda Co For Auto Parts
Spare Parts Of Cars

تقرير كشف حساب

من تاريخ : 2025/01/01 الى 2025/11/26

اسم الحساب : الصندوق العام — رقم الحساب : 1210070001

رقم المستند	نوع العملية	التاريخ	الملاحظة	المرجع	المدين	الدائن
			الرصيد الافتتاحي		0.00	0.00
700114	فاتورة مبيعات	2025/01/02	مصنع شركة شمس للرخام والجرانيت المحدودة		165.00	
700115	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		280.00	
700116	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		115.00	
700117	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		65.00	
700118	فاتورة مبيعات	2025/01/02	مؤسسة القوة الباردة للتبريد والتكييف		35.01	
700119	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		160.00	
700123	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		63.00	
700124	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		920.00	
700125	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		40.00	
700127	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		55.00	
700128	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		320.00	
700129	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		45.00	
700131	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		175.00	
700132	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		280.00	
700134	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		285.00	
700136	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		80.01	
700137	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		510.00	
700138	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		55.00	
700139	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		115.00	
700140	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		50.00	
700141	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		110.40	
700142	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		160.00	
700144	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		250.00	
700146	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		375.00	
700148	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		120.00	
700149	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		25.00	
700150	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		135.00	
700154	فاتورة مبيعات	2025/01/02	مقابل فاتورة مبيعات		690.00	
700156	فاتورة مبيعات	2025/01/02	شركة اليد الحلبا للنقل البري		335.00	
700007	سندات الصرف	2025/01/02	دفعة من الحساب نقدا			700.00
700008	سندات الصرف	2025/01/02	شركة امجاد الوفاء			500.00
700009	سندات الصرف	2025/01/02	مصاريف البوفية			120.00
700010	سندات الصرف	2025/01/02	اجور نقل بضاعه			115.00
700011	سندات الصرف	2025/01/02	خصم مسموح به			50.00
700008	قيود يومية	2025/01/02	اقفال موازنة نقط البيع			6,725.00
700010	قيود يومية	2025/01/02	اقفال الحركة اليومية			1,308.00
			اجمالي العمليات		3,872,241.02	3,872,087.19
			اجمالي الرصيد النهائي المستحق عليكم		153.83	0.00

طباعة بواسطة : باسم باصهبي — تاريخ الطباعة : 2025/11/26 09:46:15

ارقام التواصل : — صفحة رقم (4 / 491) — العنوان : جدة - طريق مكة القديم - كيلو 14

4.4 Documents Analysis

Figure 4.6 shows a complete financial statement for Al Nahda Auto Parts Company, covering the period from January 1, 2025, to November 26, 2025. Analyzing the details of the statement reveals how the company records, verifies, and tracks its financial transactions. This helps us understand how Newpoint's system is structured for invoices, payments, and account balances, and how this impacts the accuracy and reliability of the financial data in general.

The first thing we notice is that the system records each invoice with a unique document number, transaction type, date, notes, and debit and credit amounts (page 1). This structured format ensures that each financial entry can be traced back to a specific supplier, customer, or internal process, reducing the likelihood of data loss or duplication. This aligns with best practices in Enterprise Resource Planning (ERP) systems, where transparency and traceability are essential for financial control.

By reviewing the transactions on the pages, particularly the recurring entries titled "Sales Invoices" and "Sales Entry Accepted," it becomes clear that NewPoint ensures consistency in recording accounts payable and receivable. For example, page 1 displays several sequential invoice entries (700001-700041), each with corresponding amounts ranging from SAR 10 to SAR 560, demonstrating that the system handles both small and large transactions with equal accuracy. This indicates a stable financial workflow and consistent application of system rules.

The report also highlights the current balance, which remains consistent throughout the document. Pages 1-4 all display identical totals at the bottom:

- Total Debit: SAR 3,872,087.19
- Total Credit: SAR 3,872,241.02

This confirms that NewPoint processes transactions accurately and maintains proper ledger balance. The presence of a small difference labeled “Amount Due from You” of SAR 153.83 demonstrates that the system automatically calculates outstanding balances, enhancing its reliability in managing financial obligations.

Analysis of this document reveals that well-organized financial documentation facilitates workflow. Clear invoice numbering, consistent formatting, and complete transaction details minimize ambiguity and enable employees to resolve issues quickly. It also reflects the company's commitment to maintaining accurate back-end data, which is crucial for preventing errors during purchasing and payment processes.

In short, this document demonstrates that Newpoint's financial recording procedures are transparent, systematic, and reliable. Understanding how data is recorded and verified in these records provides valuable insight into how the system prevents financial errors, fosters trust with suppliers, and improves operational efficiency.

4.6 Requirements sheet

Functional Requirement	Description	Example
Process-oriented	A process the system must perform; a process the system must do	1-The system must allow employees to generate purchase orders automatically when inventory is low. 2-The system must send automated notifications to suppliers when stock is low. 3-The system must enable a unified dashboard for comparing supplier quotes and choose between them. 4-The system should allow users to filter and sort supplier interactions by date, status, and priority. 5-The system should enable users to approve or reject supplier quotations through an integrated approval workflow. 6-The system must allow users to track the status of each supplier request from initiation to completion. 7-The system must support an integrated communication channel for employees and suppliers within the platform.
Information-oriented	Information the system must contain	8- The system must display real-time inventory levels to help users make accurate purchasing decisions. 9-The system should provide a comparison view that shows suppliers' price quotes side by side. 10-The system should provide structured data outputs that support decision-making regarding supplier selection. 11-The system should display clear indicators of pending, completed, and overdue supplier tasks. 12-The system must generate detailed reports on supplier performance and response times. 13-The system should generate summary reports on user satisfaction levels for management review. 14-The system must store all supplier communication records securely for future reference. 15-The system should provide visual dashboards that highlight the most frequently selected system enhancements.

Table 4.8: Functional Requirements

Nonfunctional Requirement	Description	Example
Operational	The physical and technical environments in which the system will operate	1-The system must be compatible with any Web browser.
Performance	The speed, capacity, and reliability of the system	2-Any interaction between the user and the system should not exceed 3 seconds. 3-The system must be available 24/7 with minimal downtime for maintenance.
Security	Improve the security of the system	4-The system must enforce strong password policies (minimum 8 characters, including numbers and symbols).
Cultural and Political	Cultural and political factors and legal requirements that affect the system	5-The system must comply with Saudi Arabian laws related to data protection, financial transactions, and accounting standards. 6-The system should also comply with local tax regulations, including the Saudi VAT requirements for financial transactions.

Table 4.9: Nonfunctional Requirements

4.7 Conclusion

This chapter explored the current challenges and opportunities for improving the Newpoint accounting system at Al Nahda Auto Parts Company through an analysis of interviews, questionnaires, and document reviews. The team identified several key issues, such as manual inventory tracking, delayed purchase orders, and ineffective communication with suppliers. These challenges hinder system performance and contribute to operational inefficiencies.

A set of functional and non-functional requirements was developed to enhance system efficiency and support the company's long-term growth and sustainability.

Chapter 5: Requirement Analysis

5.1 Introduction

This section describes the current and proposed enhancements for the NewPoint Accounting System used at Al-Nahdah Auto Parts Company. The discussion starts with the existing workflow, whereby the Inventory Management System communicates with the Purchase, Sales, and Financial modules over stock levels, communication with suppliers, and updating of financial information. The proposed system will add automation in notifying suppliers, a consolidated dashboard where quotes can be compared, an enhanced purchase order process, and comprehensive reports on supplier performance. The functional and non-functional requirements are identified in the chapter, user stories based on the needs of the system are developed, and prioritized using the MoSCoW method. Finally, the Definition of Ready, Definition of Done, and Sprint 1 Backlog set out guidelines and tasks for the first development iteration.

5.2 Current system Analysis
The scenario of the current system:

AlNahda Auto Parts is a mid-sized company specializing in the distribution of automotive spare parts in the local market. The company uses the NewPoint Accounting System, which integrates several modules such as the Purchase Management System, Sales Management System, and Financial Management System. The Inventory Management System is the core component that coordinates stock movements between departments and external suppliers.

The process begins when the employee using Inventory Management System detects a shortage in stock based on warehouse levels stored in the inventory stock file. The employee sends the current stock needs to the Purchase Management System. The employee then communicates with supplier to request price quote.

Once the supplier responds with a price quote, the employee sends it to the purchase department for review. After approval is confirmed, a price quote confirmation is sent to the supplier. And A copy of the purchase order details is recorded in the purchase order file.

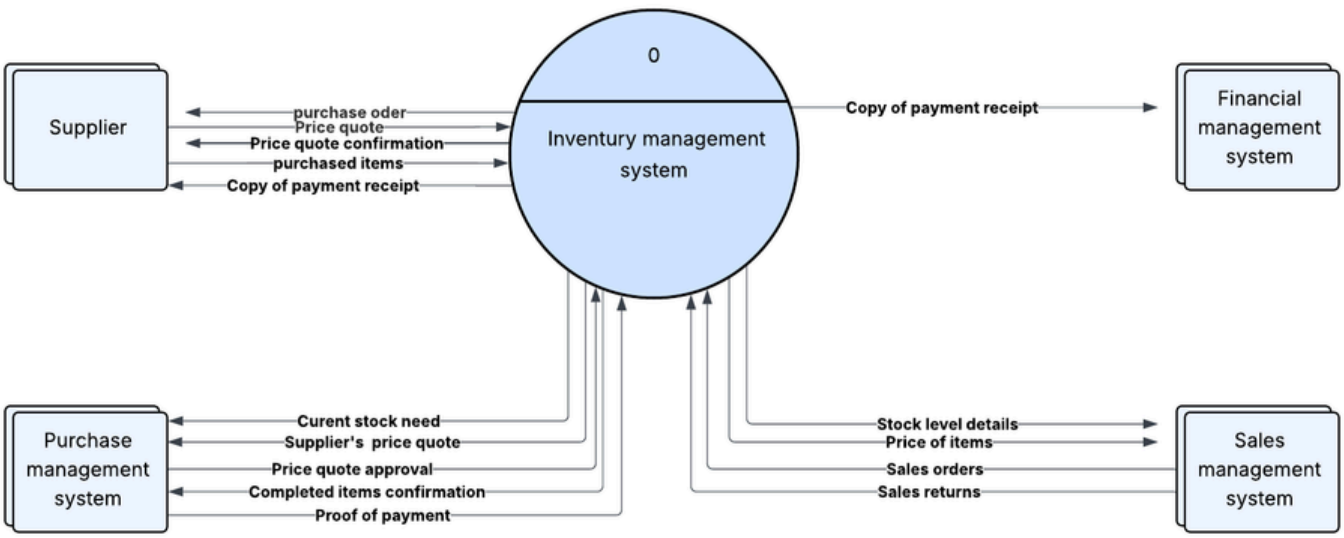
After the ordered items arrive, warehouse employees receive and verify the purchased items using the Inventory Management System. The stock levels are then updated in the inventory stock file, and a completed items confirmation is sent to the purchase department so that they could be able to pay for the suplier and send the proof of payment to the inventory system which will send it to the supplier. Another copy of the payment receipt is shared with the Financial Management System and stored in the payments file for accounting purposes.

Meanwhile, the Inventory Management System provides the Sales Management System with updated stock level details and item pricing to support customer sales orders. When a sales order is confirmed, the system automatically performs inventory deduction and updates warehouse records to ensure accurate stock availability.

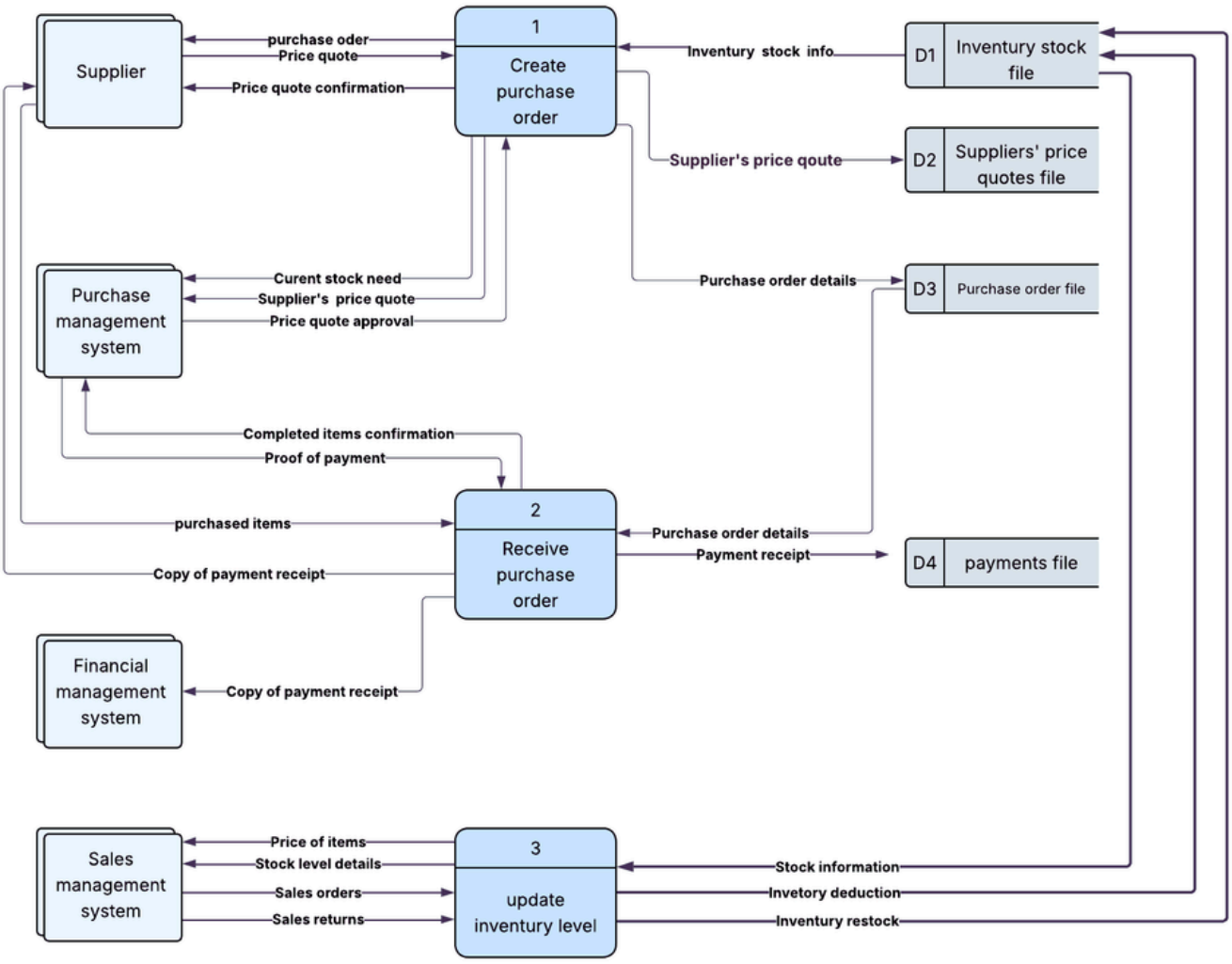
In case of sales returns, the returned items' information is sent from the Sales Management System back to the Inventory Management System, which performs inventory restock and adjusts stock file levels accordingly.

Through this fully integrated workflow, the Inventory Management System ensures efficient stock tracking, timely replenishment, and seamless communication between departments, supporting accurate business operations within AlNahda Auto Parts.

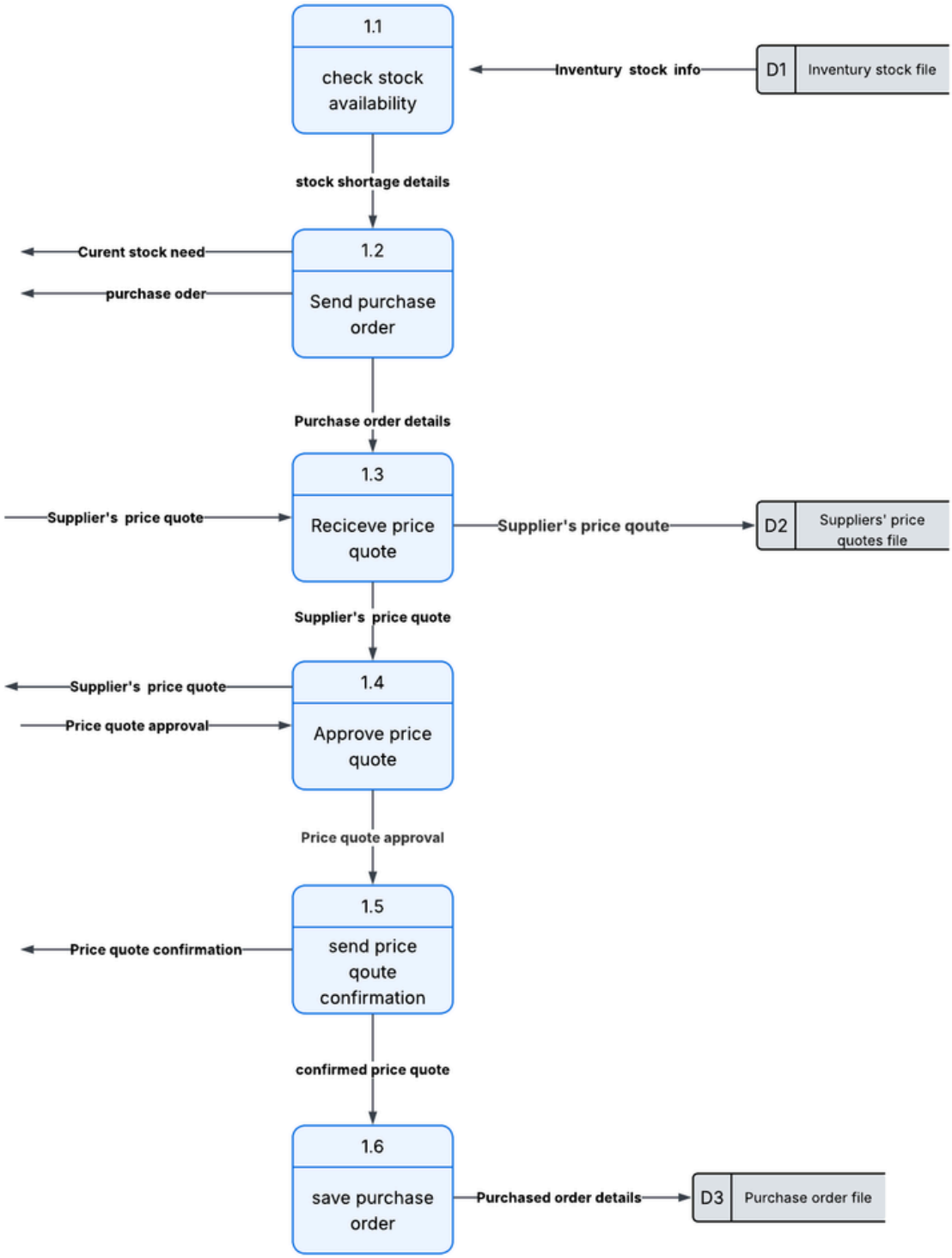
5.2.1 Current Context Diagram



5.2.2 Current DFD 0



5.2.3 Current Child Diagram



Chapter 5: Requirement Analysis

proposed System scenario :

AlNahda Auto Parts is a mid-sized company specializing in the distribution of automotive spare parts in the local market. The company uses the NewPoint Accounting System, which integrates several modules such as the Purchase Management System, Sales Management System, and Financial Management System. The Inventory Management System is the core component that coordinates stock movements between departments and external suppliers.

The process begins when the Inventory Management System detects a shortage in stock based on warehouse levels stored in the inventory stock file. The system automatically notify all suppliers that a shortage happened which make the interested of them respond by sending there price quotes, these price quotes will be shown to the purchase management system with the current stock needs, the purchase department will use a dashboard included on the system that shows all suppliers price quotes so they could easily compare between them with the help also from the suppliers information file to get information about suppliers and choose the best price quote. The confirmed price quote will be sent to the inventory system that will send the price quote approval to the selected supplier.

After selecting a supplier, the employee generates a Purchase Order (PO) through the Inventory Management System and save it in a purchase ordes file.The PO includes part details, quantities, expected delivery dates, and pricing. Throughout the process, the supplier updates the company with the current status of the order, and receives order remainder or delay notifications when necessary.

After the ordered items arrive, warehouse employees receive and verify the purchased items using the Inventory Management System. The stock levels are then updated in the inventory stock file, and a completed items confirmation is sent to the purchase department so that they could be able to pay for the suplier and send the proof of payment to the inventory system which will send it to the supplier. Another copy of the payment receipt is shared with the Financial Management System and stored in the payments file for accounting purposes.

The system also generates a Supplier Performance Report, evaluating: delivery accuracy, price competitiveness, responsiveness, quality of delivered parts. A copy of this report is sent to the suppliers file and to the purchase department and the management for decision-making regarding future supplier selection.

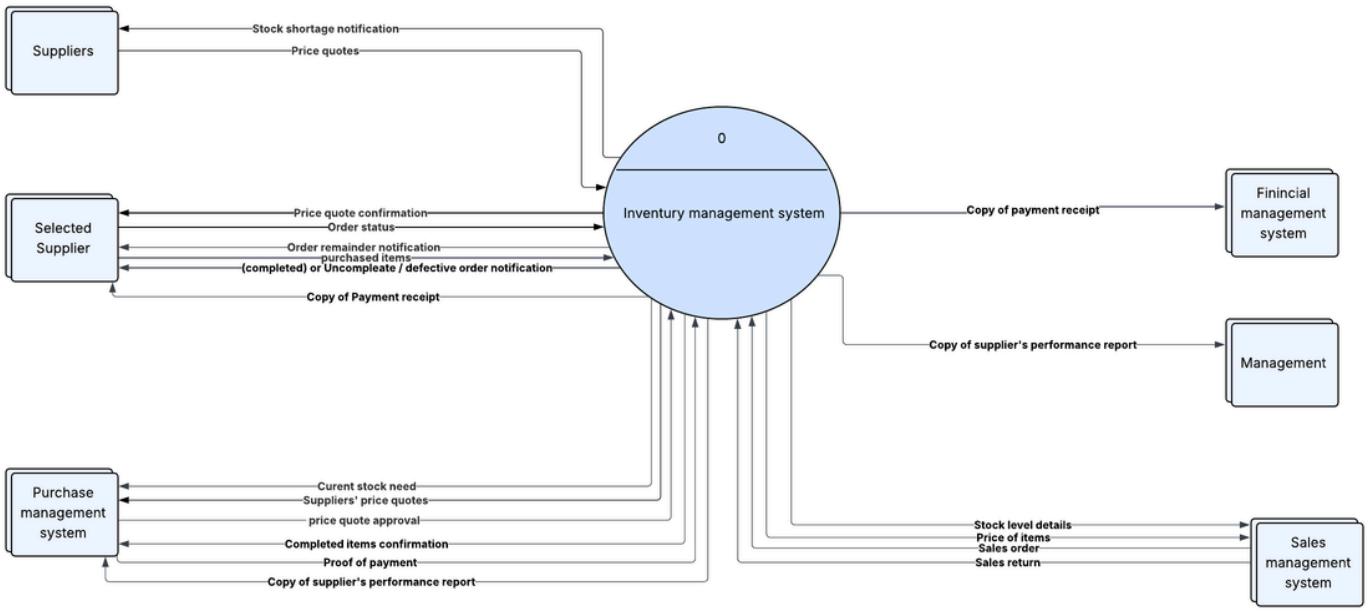
Meanwhile, the Inventory Management System provides the Sales Management System with updated stock level details and item pricing to support customer sales orders. When a sales order is confirmed, the system automatically performs inventory deduction and updates warehouse records to ensure accurate stock availability.

In case of sales returns, the returned items' information is sent from the Sales Management System back to the Inventory Management System, which performs inventory restock and adjusts stock file levels accordingly.

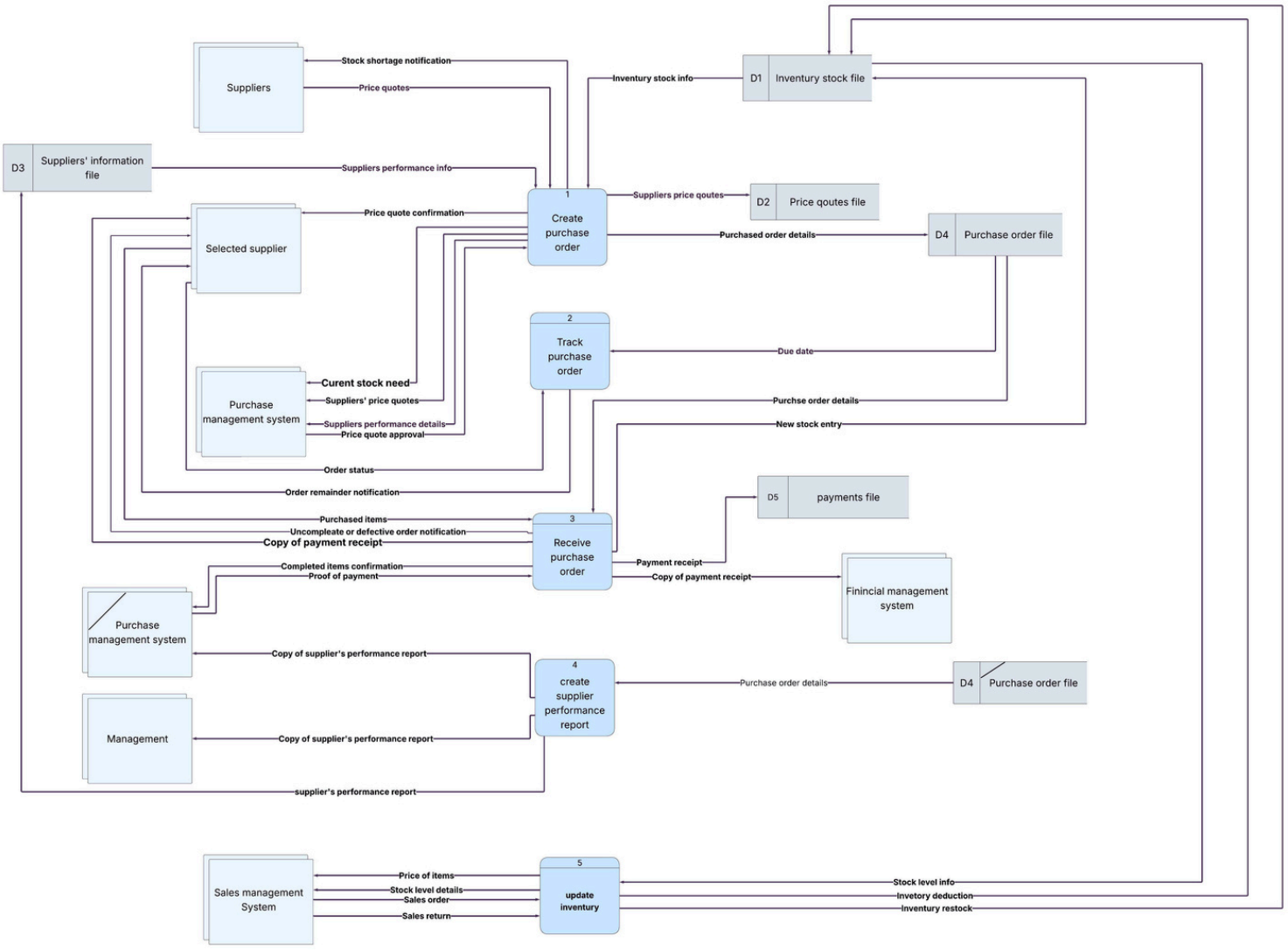
Through this fully integrated workflow, the Inventory Management System ensures efficient stock tracking, timely replenishment, and seamless communication between departments, supporting accurate business operations within AlNahda Auto Parts.

5.3 Proposed system.

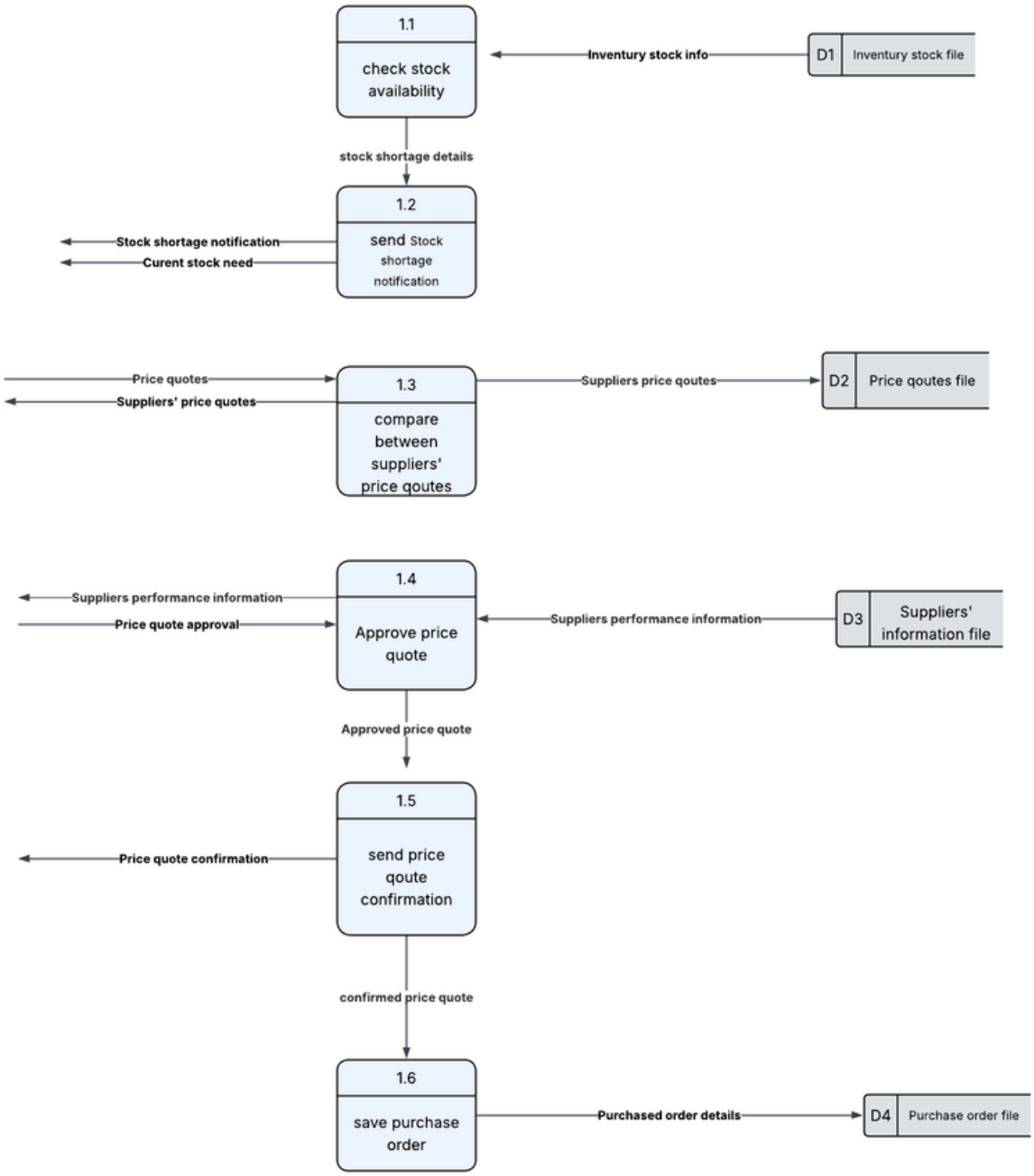
5.3.1 Proposed Context Diagram



5.3.2 Proposed DFD 0



5.3.3 Proposed Child Diagram



Chapter 5: Requirement Analysis

User stories

Functional Requirements	Examples	User Stories
Process-Oriented	1-The system must send automated notifications to suppliers when stock is low. 2-The system must enable a unified dashboard for comparing supplier quotes and choose between them. 3-The system should allow users to filter and sort supplier interactions by date, status, and priority. 4-The system should enable users to approve or reject supplier quotations through an integrated approval workflow. 5-The system must allow users to track the status of each supplier request from initiation to completion. 6-The system must support an integrated communication channel for employees and suppliers within the platform. 7- The system should send automated notification to the supplier when the oder is late.	1. As an employee, I want the system to send automated notifications to suppliers when stock is low so that orders can be placed quickly. 2. As a purchase manager, I want a unified dashboard for comparing supplier quotes so that I can choose the best option easily. 3. As an employee, I want to filter and sort supplier interactions by date, status, and priority so that I can manage supplier communication efficiently. 4. As a purchase manager, I want to approve or reject supplier quotations through an integrated workflow so that procurement decisions follow the proper process. 5. As an employee, I want to track the status of each supplier request from initiation to completion so that I can monitor progress. 6. As an employee, I want an integrated communication channel with suppliers so that all messages remain organized within the system. 7. as an employee I want the system to send automated notification to the supplier when there is a delay, So that I can ensure timely follow-up and avoid delays in inventory replenishment
Information - Oriented	8- The system must display real-time inventory levels to help users make accurate purchasing decisions. 9-The system should provide a comparison view that shows suppliers' price quotes side by side. 10-The system should provide structured data outputs that support decision-making regarding supplier selection. 11-The system should display clear indicators of pending, completed, and overdue supplier tasks. 12-The system must generate detailed reports on supplier performance and response times. 13-The system should generate summary reports on user satisfaction levels for management review. 14-The system must store all supplier communication records securely for future reference. 15-The system should provide visual dashboards that highlight the most frequently selected system enhancements.	8. As an employee, I want to view real-time inventory levels so that I can make accurate purchasing decisions. 9. As a purchase manager, I want the system to show suppliers' price quotes side-by-side so that I can easily compare prices. 10. As a purchase manager, I want structured system data outputs so that I can make informed decisions regarding supplier selection. 11. As an employee, I want the system to show clear task status indicators so that I can manage supplier-related tasks efficiently. 12. As a manager, I want detailed reports on supplier performance and response time so that I can evaluate supplier reliability. 13. As a system owner, I want summary reports on system user satisfaction so that management can assess system effectiveness. 14. As an employee, I want the system to store all supplier communication records securely so that I can reference them in the future. 15. As a system owner, I want visual dashboards showing frequently requested enhancements so that improvement planning becomes easier.

Chapter 5: Requirement Analysis

User stories

NonFunctional Requirements	Examples	User Stories
Operational	1-The system must be compatible with any Web browser.	1. As a user, I want the system to work on any web browser so that I can access it easily across different devices.
Performance	2-Any interaction between the user and the system should not exceed 3 seconds. 3-The system must be available 24/7 with minimal downtime for maintenance.	2- As a user, I want every system interaction to respond within 3 seconds so that my work is not delayed 3- As a user, I want the system to be available 24/7 with minimal downtime so that I can access it whenever I need.
Security	4-The system must enforce strong password policies (minimum 8 characters, including numbers and symbols).	4- As a user, I want the system to enforce strong password rules so that my account and data remain secure.
Cultural and Political	5-The system must comply with Saudi Arabian laws related to data protection, financial transactions, and accounting standards. 6-The system should also comply with local tax regulations, including the Saudi VAT requirements for financial transactions.	5- As an organization, I want the system to comply with Saudi data protection and financial laws so that we operate legally. 6- As an accountant, I want the system to comply with Saudi VAT rules so that financial transactions are accurate and legally correct.

5.4.1 Product Backlog

5.4.2 Priorities Requirements

5.4.2.1 MOSCOW

ID	User Story	Prioritization	sprint	Acceptance criteria	Estimate
US001	As an employee, I want the system to send automated notifications to suppliers when stock is low so that orders can be placed quickly.	Must have	1	1. The system must detect low-stock thresholds based on predefined minimum quantities. 2. A notification must be automatically sent to the assigned suppliers once the threshold is reached. 3. Employees must be able to customize low-stock thresholds for each item. 4. The system must alert the employee if an automated notification fails to send.	10
US002	As a purchase manager, I want a unified dashboard for comparing supplier quotes so that I can choose the best option easily.	Must have	1	1. The system must consolidate all supplier quotes into one unified workflow accessible from the purchase module. 2. The dashboard must allow the manager to navigate through the comparison process in a step-by-step flow (view → filter → select → finalize). 3.The dashboard must automatically update whenever new quotes are received or modified.	7
US003	As a purchase manager, I want the system to show suppliers' price quotes side-by-side so that I can easily compare prices.	Must Have	1	1. The system must display supplier quotes in a structured side-by-side table showing price, delivery time, item details, and terms. 2. All fields must be aligned and standardized to ensure clear informational comparison between suppliers. 3. Any differences in price, delivery dates, or terms must be visually highlighted to enhance readability. 4. Users must be able to download or export the comparison table as a PDF or Excel file for reference.	10
US004	As an employee, I want to track the status of each supplier request from initiation to completion so that I can monitor progress.	Must have	2	1. The system must show a clear status for every supplier request (e.g., Initiated, In Review, Approved, Sent, Delivered, Completed). 2. The status must update automatically whenever an action is performed by the employee, manager, or supplier. 3.The system must allow filtering supplier requests by status, date, or supplier name.	15
US005	as an employee I want the system to send automated notification to the supplier when there is a delay, So that I can ensure timely follow-up and avoid delays in inventory replenishment	Must have	2	1. The system must detect delays automatically based on the expected delivery date or supplier response time. 2. When a delay is detected, an automated notification must be sent to the supplier through the system (email or in-app message). 3. The system must log every delay notification sent, including timestamp and supplier name.	7

5.4.1 Product Backlog

ID	User Story	Prioritization	sprint	Acceptance criteria	Estimate
US006	As an employee, I want to view real-time inventory levels so that I can make accurate purchasing decisions.	Must have	1	1. The system must display inventory levels that update automatically without page refresh. 2. The employee must be able to view stock quantities for all items in a single dashboard. 3. The displayed inventory data must not be older than 60 seconds. 4. The system must show an alert if data cannot be refreshed or synced.	5
US007	As a user, I want the system to work on any web browser so that I can access it easily across different devices.	Must have	1	1. The system must run smoothly on Chrome, Safari, Firefox, and Edge. 2. Core functionalities must behave consistently across all supported browsers. 3. The interface must be responsive on different screen sizes (desktop, tablet, mobile). 4. No browser-specific errors should appear when accessing any feature.	15
US008	As an organization, I want the system to comply with Saudi data protection and financial laws so that we operate legally.	Must have	2	1. The system must store and process all user data in accordance with Saudi Personal Data Protection Law (PDPL) requirements. 2. Financial records must be retained and archived following ZATCA/Saudi financial regulations for audit purposes. 3. The system must restrict access to sensitive data based on role-based permissions as required by PDPL. 4. All data transmissions must be encrypted (e.g., TLS/HTTPS) to ensure compliance with data security standards.	10
US009	As an accountant, I want the system to comply with Saudi VAT rules so that financial transactions are accurate and legally correct.	Must have	2	1. The system must automatically calculate VAT at the correct rate (e.g., 15%) according to Saudi VAT laws. 2. All invoices must include VAT-required fields: VAT number, taxable amount, VAT amount, total amount. 3. The system must prevent saving or posting any transaction that does not meet VAT calculation requirements.	5
US010	As a user, I want every system interaction to respond within 3 seconds so that my work is not delaye d	Must have	3	1. All user-facing pages must load in 3 seconds or less under normal operating conditions. 2. Core features (search, saving records, loading dashboards) must respond within 3 seconds. 3. The system must maintain the response-time requirement even during periods of high user traffic. 4. If an action exceeds 3 seconds, a loading indicator or progress message must appear to inform the user.	20

Chapter 5: Requirement Analysis

ID	User Story	Priority	sprint	Acceptance criteria	Estimate
US011	As a purchase manager, I want to approve or reject supplier quotations through an integrated workflow so that procurement decisions follow the proper process.	Must have	3	1. The system must provide a dedicated approval workflow where the manager can approve or reject quotations. 2. Quotations must move through defined workflow stages (submitted → under review → approved/rejected). 3. The manager must be able to add comments or notes when approving or rejecting a quotation. 4. The system must notify the requester when a quotation is approved or rejected.	10
US012	As an employee, I want an integrated communication channel with suppliers so that all messages remain organized within the system .	Should have	3	1. The system must provide an in-app messaging interface for communication with suppliers. 2. All messages must be stored and linked to the related purchase order or supplier request. 3. The system must support sending attachments such as PDFs, quotations, or documents. 4. The employee must receive notifications when suppliers send new messages.	15
US013	As a purchase manager, I want structured system data outputs so that I can make informed decisions regarding supplier selection .	Should have	5	1. The system must generate structured data outputs that include price, delivery time, quality metrics, and supplier rating. 2. All outputs must follow a standardized format for consistency (e.g., tabular, labeled fields, KPIs). 3. The manager must be able to export the structured data in PDF or Excel format.	10
US0014	As a business manager, I want detailed reports on supplier performance and response time so that I can evaluate supplier reliability .	Should have	2	1. Reports must include key performance indicators such as response time, delivery punctuality, price accuracy, and quote acceptance rate. 2. The system must allow generating reports for a specific supplier or a date range. 3. Performance data must be displayed in tables and charts for better readability.	10
US015	As a user, I want the system to be available 24/7 with minimal downtime so that I can access it whenever I need.	Should have	4	1. System uptime must meet or exceed 99.5% availability per month. 2. Scheduled maintenance must be announced to users at least 24 hours in advance. 3. The system must automatically fail over to a backup server if the main server is down. 4. Users must be able to access all core features at any time without service interruption.	20

Chapter 5: Requirement Analysis

ID	User Story	Priority	sprint	Acceptance criteria	Estimate
US016	As a user, I want the system to enforce strong password rules so that my account and data remain secure	Should have	3	1. The system must require passwords to include uppercase, lowercase, numbers, and special characters. 2. Passwords must be at least 8–12 characters long depending on system policy. 3. The system must prevent users from reusing the last 5 previously used passwords.	5
US017	As an employee, I want the system to show clear task status indicators so that I can manage supplier-related tasks efficiently.	Should have	4	1. The system must display task statuses using standardized labels (e.g., Pending, In Progress, Waiting for Supplier, Completed). 2. Each task must include status-specific icons or color indicators for quick identification. 3. Task status must update automatically when progress is recorded.	10
US018	As an employee, I want to filter and sort supplier interactions by date, status, and priority so that I can manage supplier communication efficiently.	Could have	4	1. The system must allow filtering interactions by date range, communication status, and priority level. 2. The system must support sorting by newest/oldest, highest priority, or unresolved interactions. 3. Filters must apply instantly and update the view without requiring a page refresh.	7
US0019	As a system owner, I want visual dashboards showing frequently requested enhancements so that improvement planning becomes easier.	Could have	5	1. The dashboard must display enhancement requests grouped by frequency and category. 2. Data must be shown with visual elements such as charts, graphs, or heatmaps. 3. The dashboard must update in real time as new enhancement requests are submitted.	15
US020	As an employee, I want the system to store all supplier communication records securely so that I can reference them in the future	Could have	5	1. All communication records must be stored in an encrypted database. 2. Communication logs must be linked to the corresponding supplier and transaction. 3. The system must prevent deletion or modification of communication history unless authorized.	5
US021	As a system owner, I want summary reports on system user satisfaction so that management can assess system effectiveness	Won't Have	5	1. The system must generate summary reports showing user satisfaction scores, trends, and key pain points. 2. Reports must aggregate data from surveys, feedback forms, and system usage patterns. 3. Reports must include visual components such as bar charts or satisfaction trend lines.	5

5.4.3 Definition Of Ready (DOR)

- ☑ AS A (someone) , I WANT TO (do something) , SO THAT(result or benefit)
- ☑ Has Acceptance Criteria that can be tested objectively
- ☑ Socialized with the entire team
- ☑ Estimated by the entire development team
- ☑ Has the right size that can be completed within a Sprint
- ☑ User Stories are INVEST

5.4.4 Definition Of Done (DOD)

- [x] Acceptance defined criteria for each user story
- [x] Unit tests written and passed
- [x] Code compiles with no errors and no warnings
- [x] New code doesn't break existing code
- [x] Test case review(Dev to review test case written)
- [x] Comments in code
- [x] Error codes added
- [x] Code reviewed by peer
- [x] Test approves user story
- [x] Acceptance criteria verified complete
- [x] Story demonstrated to product owner and accepted on target platform

5.5 Sprint 1 Backlog

Task ID	User Stories	Acceptance Criteria	priority	Estimate
1	As an employee, I want to view real-time inventory levels so that I can make accurate purchasing decisions.	1. The system must display inventory levels that update automatically without page refresh. 2. The employee must be able to view stock quantities for all items in a single dashboard. 3. The displayed inventory data must not be older than 60 seconds. 4. The system must show an alert if data cannot be refreshed or synced.	Must	5
2	As an employee, I want the system to send automated notifications to suppliers when stock is low so that orders can be placed quickly.	1. The system must detect low-stock thresholds based on predefined minimum quantities. 2. A notification must be automatically sent to the assigned suppliers once the threshold is reached. 3. Employees must be able to customize low-stock thresholds for each item. 4. The system must alert the employee if an automated notification fails to send.	Must	10
3	As a purchase manager, I want a unified dashboard for comparing supplier quotes so that I can choose the best option easily.	1. The system must consolidate all supplier quotes into one unified workflow accessible from the purchase module. 2. The dashboard must allow the manager to navigate through the comparison process in a step-by-step flow (view → filter → select → finalize). 3.The dashboard must automatically update whenever new quotes are received or modified.	Must	7
4	As a purchase manager, I want the system to show suppliers' price quotes side-by-side so that I can easily compare prices.	1. The system must display supplier quotes in a structured side-by-side table showing price, delivery time, item details, and terms. 2. All fields must be aligned and standardized to ensure clear informational comparison between suppliers. 3. Any differences in price, delivery dates, or terms must be visually highlighted to enhance readability. 4. Users must be able to download or export the comparison table as a PDF or Excel file for reference.	Must	10
5	As a user, I want the system to work on any web browser so that I can access it easily across different devices.	1. The system must run smoothly on Chrome, Safari, Firefox, and Edge. 2. Core functionalities must behave consistently across all supported browsers. 3. The interface must be responsive on different screen sizes (desktop, tablet, mobile). 4. No browser-specific errors should appear when accessing any feature.	Must	15

5.5 Sprint 1 Backlog

The information above illustrates the current efficiencies and how it will greatly enhance productivity with the proposed NewPoint Accounting System improvements. Automation of suppliers, inventories, and reporting will enhance the operation and reduce manual effort. Identified requirements, user stories, priorities, DoR, DoD, and Sprint 1 plan form a solid framework for productive system development. Overall, these improvements will help Al-Nahdah Auto Parts achieve more accurate, responsive, and integrated accounting and inventory processes.

Chapter 5: Requirement Analysis

Velocity-50 points

Product backlog

1. As an employee, I want the system to send automated notifications to suppliers when stock is low so that orders can be placed quickly. **10pts**
2. As a purchase manager, I want a unified dashboard for comparing supplier quotes so that I can choose the best option easily. **7pts**
3. As an employee, I want to filter and sort supplier interactions by date, status, and priority so that I can manage supplier communication efficiently.**7pts**
4. As a manager, I want to approve or reject supplier quotations through an integrated workflow so that procurement decisions follow the proper process.**10pts**
5. As an employee, I want to track the status of each supplier request from initiation to completion so that I can monitor progress.**15pts**
6. As an employee, I want an integrated communication channel with suppliers so that all messages remain organized within the system.**15pts**
7. as an employee I want the system to send automated notification to the supplier when there is a delay, So that I can ensure timely follow-up and avoid delays in inventory replenishment.**7pts**

8. As an employee, I want to view real-time inventory levels so that I can make accurate purchasing decisions. **5pts**

9. As a purchase manager, I want the system to show suppliers' price quotes side-by-side so that I can easily compare prices.**10pts**
10. As a purchase manager, I want structured system data outputs so that I can make informed decisions regarding supplier selection.**10pts**
11. As an employee, I want the system to show clear task status indicators so that I can manage supplier-related tasks efficiently.**10pts**
12. As a manager, I want detailed reports on supplier performance and response time so that I can evaluate supplier reliability. **10pts**
13. As a system owner, I want summary reports on system user satisfaction so that management can assess system effectiveness. **15pts**
14. As an employee, I want the system to store all supplier communication records securely so that I can reference them in the future.**5 pts**
15. As a system owner, I want visual dashboards showing frequently requested enhancements so that improvement planning becomes easier.**15pts**

16-As a user, I want the system to work on any web browser so that I can access it easily across different devices.
17- As a user, I want every system interaction to respond within 3 seconds so that my work is not delayed **20pts**
18- As a user, I want the system to be available 24/7 with minimal downtime so that I can access it whenever I need.**20pts**
19- As an organization, I want the system to comply with Saudi data protection and financial laws so that we operate legally. **10pts**
20- As an accountant, I want the system to comply with Saudi VAT rules so that financial transactions are accurate and legally correct.**5pts**

Sprint commitment

As an employee, I want to track the status of each supplier request from initiation to completion so that I can monitor progress.**15pts**

as an employee I want the system to send automated notification to the supplier when there is a delay, So that I can ensure timely follow-up and avoid delays in inventory replenishment.**7pts**

As an organization, I want the system to comply with Saudi data protection and financial laws so that we operate legally. **10pts**

As an accountant, I want the system to comply with Saudi VAT rules so that financial transactions are accurate and legally correct. **5pts**

As a business manager, I want detailed reports on supplier performance and response time so that I can evaluate supplier reliability. **10pts**

Sprint backlog

1. Tracking Supplier Request Status (15 pts)

Design request status workflow

Implement status update functions

Display status timeline to users

Test end-to-end tracking

2. Automated Supplier Delay Notifications (7 pts)

Define delay conditions

Implement notification trigger

Send automated supplier alerts

Test notification accuracy

3. Compliance with Saudi Data & Financial Laws (10 pts)

Identify required legal standards

Implement data protection controls

Apply financial compliance rules

Validate compliance through testing

4. Compliance with Saudi VAT Rules (5 pts)

Identify VAT calculation requirements

Implement VAT logic in transactions

Validate VAT fields in invoices

Test VAT accuracy

5. Supplier Performance & Response Time Reports (10 pts)

Collect supplier performance data

Build reporting dashboards

Implement response-time metrics

Test report accuracy

Total team velocity = 50 Story Points
Total team capacity = 15 members working 6 days a week, 8 hours a day
Sprint duration = 1 month (approximately 24 working days)

5.6 Conclusion

In conclusion, the current and proposed systems for AlNahda Auto Parts demonstrates how automation and integration can significantly improve operational efficiency. By enhancing supplier communication, streamlining purchasing and inventory processes, and introducing real-time data, the proposed NewPoint Accounting System provides a stronger, faster, and more accurate workflow. The detailed functional and non-functional requirements, user stories, DoR, DoD, and sprint planning form a solid foundation for developing a reliable and compliant system. Overall, these improvements will support better decision-making, reduce manual effort, and ensure more accurate and responsive inventory and accounting operations for the company.

Resources

نيو بوينت من افضل واسهل البرامج المحاسبية ونقاط البيع. (n.d.). /https://newpointerp.com

Direct communication with Al Nahda Company was conducted to gather information.